

CS482/495/496 Software Project Proposal:

add your tentative project title here

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1 Client Information

By sharing this client information and the rest of this document, you are stating that this client has provided this project as something they want (not something you created and asked if they wanted), and that they are interested in having you complete this project for your capstone.

- Client name: Prof. Ebert
- Client title: Professor
- Client email address: eebert@loyola.edu
- Client employer: University Of Loyola Maryland
- How you know the client: Professor at Loyola

2 Project Description

2.1 Overview

[Add a few paragraphs describing your project succinctly. What problem are you trying to solve, what is the purpose of your project? Why does your client want this project?]

2.2 Key Features

[At this point you should have a basic understanding of your client's needs. List out the key features of the software system the client wants you to build.]

2.3 Why this Project is Interesting

[Why did you decide this project was interesting enough to you to be a capstone project? What about this project is enticing? Why should anyone care?]

2.4 Areas of CS required

[What subfields of computer science seem most likely to be relevant to your project? A capstone must involve multiple.]

2.5 Potential Concerns and Questions

[Is there any aspect of this project that makes you unsure if it will work, either due to your own interests/background, or that you aren't sure if it fits the requirements? Are there questions you have about this project that you want instructor feedback about?]

2.6 Summary of Efforts to Find a Project

(Not necessary for 482) [Briefly list out when/how you’ve discussed with this client, and if you’ve discussed with other clients who either didn’t work out or didn’t respond. If you considered a different project and it didn’t work out, why didn’t it work out?]

[Most CS495 projects end here. The sections below are for CS482 and CS496 software projects].

2.7 Comparison to Draft

[For CS496 only, focus on highlighting the major differences between the draft proposal in CS495 and this one here. If there are no major differences, you can remove this subsection.]

3 Requirements

3.1 Non-Functional Requirements

[Non-functional requirements are just as important as functional requirements. Dont forget to specify them.]

ID	NFR Title	Category	Description
NFR1	NFR Example 1	Usability	Description of the NFR (it does not follow a user story template)
NFR2	NFR Example 2	Security	Description of the NFR (it does not follow a user story template)

Table 1: Non-Functional requirements

3.2 Functional Requirements (User Stories)

[In CS482, all functional requirements are written as User Stories. In CS496, some projects may use a different template to write the requirements. The table below is an example of writing the Stories. Adapt accordingly to different templates or if you want to display more info.]

ID	Story Title	Points	Description
S1	Story Example 1	5	As a user, I want to write a user story example, so that people will understand them.
S2	Story Example 2	2	As a user, I want to write a user story example, so that people will understand them.

Table 2: Functional requirements as User Stories.

4 System Design

4.1 Architecture

[Which type of software architecture are you team following? Layered architecture, MVC, other? What are the main modules for your software?]

Our team has chosen to utilize the MVC structured architecture to allow for obvious focus point on our development. Using MVC we can split up our work into 3 main categories: Model, View, and Controller. Using this style of structure will promote scalability, mendability, and maintenance as we move further down the road. It will allow for developers to handle different parts of the project, avoiding the conflict of waiting on another group member to finish before starting your work. When it comes to tools we will use for each part of MVC that will be explained in our Technology section. The module will be the main branch that handles our data that will contain important information for the game consistently and securely. The controller is

the link between the model and the view. It will take in requests data from the user and retrieve data from the model to display on the view. The view is the display of the structure and it allows for the user to see the information our software is displaying. Many benefits will be created from using this structure that will allow for efficiency, debugging, organization, modular development, team collaboration, and maintainable software.

4.2 Diagrams

[CS482, on sprints/iterations 2-3, you need to create and update a diagram (check the assignment for which type of diagram). On CS496, since before sprint/iteration 1 you should have a class diagram and keep it up-to-date.]

4.3 Technology

For the front-end, we will use React to handle with the user interface. In terms of our back-end we will use JavaScript to handle the interaction of the front-end with data. For API usage, we are going to utilize NodeJS to gather the necessary information to display on our website.

4.4 Coding Standards

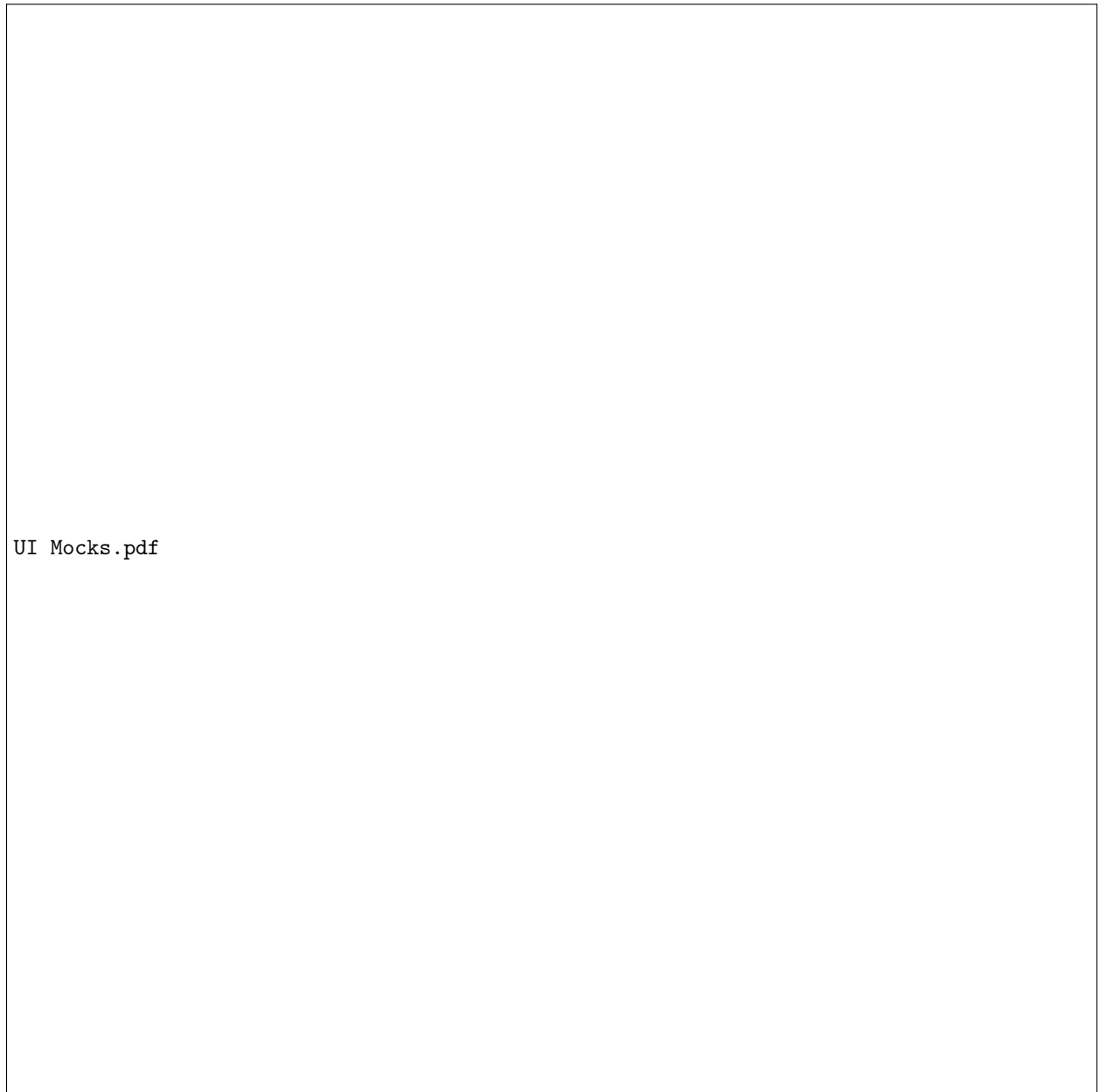
-We are going to limit the use of global variables. -Header Format of each different Module: Name Date Author Synopsis Variables -We are going to utilize CamelCase for Variables and Functions -Final Code Shouldn't be pushed unless tested beforehand or explicitly given notice of an issue -Comments must be made in order to help entire team understand code

4.5 Data

Logical Model of the database entities and their relationships:

```
Adult ( id_Adult; Name; Dob; Email; Phone; Permissions; );
Player/Child ( @id_player; #id_Adult; #id_Team; Dob; );
Coach ( @id_Coach, #id_Adult; #id_Team; );
Team ( @id_Team; #id_Coach; #id_Manager; Playerlist;
);
Guest ( @id_Guest; Permissions; );
Admin ( @id_Admin; #id_Adult; );
Manager ( @id_Manager; #id_Team; #id_Adult; );
Parent ( @id_Parent; #id_Adult; );
```

4.6 UI Mocks



UI Mocks.pdf

UI Mocks.pdf

5 Iterations

5.1 Iteration Planning

[In CS496, you plan all iterations beforehand. In CS482, you update the planning here at each iteration.]

5.2 Iteration/Sprint 1

5.2.1 Planning

Will: P0 Establishing Database 1pt. U1 Rate my manager 1pt. Manager invites coach 1pt. Manager Invites player 1pt. Zoe: U11 Landing page 2pts. G0 Guest Page2pts Fabri: U3 Login/Guest 1pt, U2 Create Scoreboard 1pt, U3 standings 1 pt, u6Color scheme 1pt Dylan: u10 Create team 1pt, ADM5 Admin Page 2 pts, ADM1 Season Lockings 1 pt

Iteration	Dates	Stories	Points
1	01/01 - 02/01	S1 Story Example, S2 Story Example 2	07
2	02/01 - 03/01	S3 Story Title, S4 Story Title, S5 Story Title, S6 Story Title	17
3	03/01 - 04/01	S7 Story Title, S8 Story Title, S9 Story Title, S10 Story Title, S11 Story Title	21
4	04/01 - 05/01	S12 Story Title, S13 Story Title, S14 Story Title, S15 Story Title	19
5	05/01 - 06/01	S16 Story Title, S17 Story Title	06
Total:			70

Table 3: Iteration Planning for Incremental Deliveries

Altered Stories Will: Dylan: Create a Team, View Team (4 pts) Fabri: Created a game, Update Game(Game Status and Score), Save Game(CRUD =4pts). Explored the use of react, while not implemented in terms of the Game method, We still have a basis template to then implement with our html pages. [Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.2.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

The stories we completed on this iteration were

Dylan - create a Team and view Teams

Will-

Zoe-

Fabri- create Game Functionality

Partially completed stories

Dylan - Using React

Will -

Zoe -

Fabri - React front end Implementation

In detail

Dylan- Created a website that creates a team using the field's name, manager, and logo. The website then uses the controller and DAO for team to store the information in the database. The website can also display all the teams that are in the database by using the controller and dao.

Will -

Zoe -

Fabri - Implemented full game functionality including creating new games, updating existing game information(status and score), and saving completed games covering all core CRUD operations(4 pts). Additionally, we explored integrating React for potential frontend expansion. While React was not directly implemented for the Game module at this stage, we successfully established a foundational template that can be connected to our existing HTML pages in future iterations. This groundwork ensures an easier transition to a more dynamic, component-based user interface later on.

5.2.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

File	% Stats	% Branch	% Funcs	% Lines	Uncovered Line #s
all files	91.16	72.22	86.48	91.66	
team-MPQ-C543Fabrizio-G-Zoe-W-Mill-T-Dylan-	98.0	50	100	100	
WSConnection.js	98.0	50	100	100	5
team-MPQ-C543Fabrizio-G-Zoe-W-Mill-T-Dylan-/Docs/Baseball/Controller	98.33	91.66	100	98.33	
teamController.js	97.87	83.33	100	97.87	93
teamController.js	100	100	100	100	
team-MPQ-C543Fabrizio-G-Zoe-W-Mill-T-Dylan-/Docs/Baseball/Model	81.52	75	75.38	81.52	20,24-25,29-30,34-35,39-40,48-52
UserDao.js	50	0	20.57	50	
game.js	100	100	100	100	
team.js	98.33	50	100	98.33	65
Test Suites: 1 failed, 4 passed, 5 total					
Tests: 2 failed, 42 passed, 45 total					
Snapshots: 0 total					
Time: 11.221 s					
all tests					

Figure 1: Coverage report

The overall coverage is pretty good at around 91, which means most of the code is being tested. The controllers and DAOs have high coverage, but UserDao.js is still low since some of its database and error-handling parts aren't tested yet. The branch coverage is also lower because we didn't include admin or error checks in this version. To improve, we'd add more tests for different CRUD cases and edge scenarios. For now, this level of coverage is solid and shows that most of the main functionality works as expected.

5.2.4 Retroerspective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).] The challenges we faced initially began with our user stories. We found them to be too vague/general so when it came time to our first sprint, each story we had thought to be rather simple, turned out to be worth more crud pts. In order to avoid the same mistake we should make sure that the user stories we take on have a clear narrative and point worth.

Dylan - Something that I learned is that it is important to have a strong and correct structure of MVC to make sure everything is easy to maintain and change. Also React is very new to me and I need to make sure I get a good understanding, so implementation will be easier.

Fabrizio - I'm currently in the process of learning how to effectively integrate React with backend database operations, specifically focusing on CRUD functionality. This experience has been valuable for helping me understand how data flows between the frontend and backend, along with how user interactions can trigger database updates. Working directly with database logic has also improved my confidence in handling database work. Overall, I'm gaining a stronger grasp of fullstack development and how each layer contributes to building interactive applications.

5.3 Iteration/Sprint 2

5.3.1 Planning

Fabrizio: C0 [1], ADM1 [1], ADM2 [1], U1 [1]. Each of these work on the backend part, but not the front end. I chose this for my Iteration 2 planning as it will complete the CRUD for Games. Total Story Points = 4. Following Will, once I've completed these I will work on other user stories.

Will: ADM5 [2], ADM4 [2], ADM3 [1]. ADM4 was partially completed, so only 1 point left for that. I chose this for my iteration as it will complete the CRUD for the Users. Total Story Points = 5. Depending on when I complete this, I may complete/work on more user stories.

Dylan: M3 [2], ADM9 [2], M9 [1] (not sure if counts as a point), M1 [1]. M9 will need the implementation of adding a file (photo) for the team logo when creating the team. It also needs a color scheme update. M1 will just need to display that logo when showing the teams. M3 will be inviting a coach to a team, so it needs to read off the coaches and have an invite button to send a message. ADM9 gives the admin the ability to delete a team if needed. Total points = 5. If this is completed and I have more time, I will invite a player system.

Zoe: U3 [2], U2 [2], M2 [2].

5.3.2 Work Done

Fabrizio: During this sprint I worked on manipulating the front end aspect of the web application for viewing and updating games. During the previous sprint I had fully completed the back end side of the

CRUD operations for utilizing these features, but didn't fully completed the front end in order to perform all functionality. I also Worked on some minor tweaks in terms of utilization as a user such as navigation and overall visual appeal.

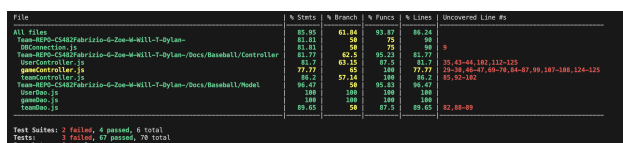
Dylan: For this iteration I made sure that I fixed my problem from last iteration. The problem was that when I made a team it did not insert an image for the team. Once that was fixed I finished the rest of my full crud for team by creating ways to update and delete teams from the data base. I also edited the color scheme to black and orange to match the wanted colors for this project.

Will: For this iteration, I focused on completing the CRUD for users and making viable html pages to interact with these functions. I finished up delete from the last iteration and finished the read and update portion.

Zoe: During this iteration I focused primarily on enhancing the user interface and integrating new backend functionality that connects users to their related features. I completely redesigned the landing page by adding proper redirection and navigation for most of the buttons. I also added a dynamic calendar that displays games by day once they have been completed. The landing page is still a work in progress since more functionality will be connected here as features are implemented. On the backend, I spent considerable time developing the ability for parents to add a child. This required significant updates to the controller and DAO layers. Once a parent signs in, they are now able to register a child account under their profile. Additionally, I implemented age restrictions to ensure that only eligible users can sign in. I also completed story M2, allowing managers to add existing players to a team through the "edit team" page.

5.3.3 Testing Coverage

A lot of the tesing was completed during our last sprint, however userDao was low so we focused on getting all of it done. More testing for frontend will be needed later on but this testing is for backend.



File	% Stmt	% Branch	% Funcs	% Lines	Uncovered Line #s
all files	80.85	63.64	83.97	88.54	
team-DAO-C482fabrizio-0-Zoe-M-Will-T-Dylan-	81.01	56	75	88.54	
gameController.js	81.01	56	75	88	
team-DAO-C482fabrizio-0-Zoe-M-Will-T-Dylan-/Docs/Baseball/Controller	81.77	62.15	95.23	81.77	
gameController.js	81.77	62.15	95.23	81.77	25, 32-44, 100, 113-125
gameController.js	77.77	55	100	77.77	28-34, 46-47, 100-79, 84-87, 99, 107-108, 124-125
team-DAO-C482fabrizio-0-Zoe-M-Will-T-Dylan-/Docs/Baseball/Model	96.47	57.14	100	96.47	55, 92-102
team-DAO-C482fabrizio-0-Zoe-M-Will-T-Dylan-/Docs/Baseball/Model	100	100	100	100	
gameDAO.js	100	100	100	100	
teamDAO.js	89.05	58	87.2	89.05	82, 85-89

Test Suites: 2 failed, 4 passed, 6 total
 Tests: 3 failed, 67 passed, 70 total
 Snapshots: 0 total

Figure 2: Coverage report

5.3.4 Retroespective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

Fabrizio: Throughout this iteration I didn't come across any major challenges in terms of working with the basic HTML front end, although React has become a bigger challenge than originally anticipated, that being said this week is a much lighter week in terms of course load so I will be able to dedicate more time to trying to implement react. During this iteration I learned how the simplest features found in common website are commonly over looked, So this week I plan on reviewing my two pages and creating a checklist of Requirements in terms of Prof. Ebert's demands and adding other features that were overlooked and implement them.

Dylan: A major challenge I face was how to implement logos into to team in our data base. I originally just had a local file that used multer to store the logos. It worked but only on my local device, and also would not have live updates unless you git pushed every time a team was made. So I was able to use a buffer which is a part of mongo db. Defintly should have started off with this but after some tweaks I got it working and should preform well on a live server. I think for the next iteration I should fully explore what mongo db has to offer before searching it up. I think that this iteration didn't contain much workload but definitely was tedious and can be avoided by planning. It made me learn a lot about images and how they are stored in database and displayed on front end. Defintly my favorite sprint as I saw a lot of progress in this iteration and can really start to see the project come together.

Will: An issue I faced during this was the update user portion. The reason this was so hard was that I ran into an issue revolving reading names. I was unsure of why this occurred but realized I changed a few lines of code in userDao and userController when trying to figure out how to finish the update function. This iteration was a lot easier compared to Iteration 1 but still had its challenges with constantly having to look back at previous code and question why it isn't working now but was working previously.

Zoe: This iteration went pretty well overall, but I did run into some small problems trying to get the add child feature working correctly. It took a while to figure out how to connect everything between the front end and the backend, especially making sure the routes and database were all lined up. I also spent time fixing small things on the landing page and making it easier to navigate, which was fun but a little time-consuming. I learned a lot more about how the backend and frontend work together and how small details can cause big issues if something doesn't match up. Next time I want to plan out my routes and data flow earlier so I don't have to spend as much time troubleshooting later.

5.4 Iteration/Sprint 3

5.4.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.4.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.4.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.4.4 Retroerspective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.5 Iteration/Sprint 4

[CS496 has 5 sprints. CS482 only has only 3 sprints (remove Iterations 4 and 5 from this doc if you are writing a doc for 482)]

5.5.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.5.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.5.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.5.4 Retrospective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

5.6 Iteration/Sprint 5

5.6.1 Planning

[Which stories did you plan for this iteration/sprint. Add the total points for this plan. You can also explain the reason behind your planning, and what major feature(s) your team is focusing on delivering by completing these stories. You may use a table for a summary display of the planning, but elaborate in text more detail in your focus and feature plan.]

5.6.2 Work Done

[Which stories did you complete in this iteration/sprint. Which ones did you partially complete? Who worked on which story? You may elaborate in paragraph(s) to add more detail about the work done.]

5.6.3 Testing Coverage

[Testing is very important. Show your coverage here. Is this coverage good enough? Explain why you think so. Is it not good enough? Explain a plan to increase the coverage. You may also elaborate on why some artifacts do not undergo much testing. If the testing changed from the last iteration, explain the reasons.]

5.6.4 Retrospective & Reflection

[What were the pitfalls, challenges, and issues you had in this iteration? How can you address them to improve the process in the next iteration? Did anything not go according to plan? Why so and how to avoid the same mistake? Write a personal reflection on what you learned in this iteration (even if a small technical thing like Database storage).]

6 Final Remarks

6.1 Overall Progress

[Have you completed everything? If so, present evidence on how you brought value to your client, and the overall client satisfaction. Otherwise, estimate how much progress you done and how long it would take to finish this project.]

6.2 Project Reflection

[Your personal reflection on the project. What lessons did you learned. What would you have done differently. How can you do better work in future projects? You may write this as a team or per person (or both)]

Appendix

[Appendix section if needed]